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Subject: IDS Roll No.: 32 Exp. No.:
Name of the Experiment:

Performed on:
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10	

Q1) Define an algorithm. List any four characteristics of an algorithm.

Ans) An algorithm is a finite sequence of well defined, step by step instruction used to solve a specific problem or perform a computation.

- Characteristics :-

i) Input :→ an algorithm takes zero or more inputs
ii) Output :→ it produces at least one put
iii) Definiteness :→ each step is clear, precise, unambiguous.
iv) Finiteness :→ the algorithm terminates after a finite number of steps.

Q2) Define data structure and abstract data type (ADT) give one example of each.

Ans) - A data structure is a way of organizing, storing and managing data in memory as that it can be accessed and modified efficiently.

- An abstract data type (ADT) is a logical description of a data type that specifies the operation that can

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be performed on it without specifying how these operations are implemented.

Examples :- Data structure : array
ADT : stack (operations : push, pop, peek)

Q3) Explain the process of converting a problem into a program. Illustrate the role of algorithms in this proce

Ans) i) Problem definition :- understand requirements and constraints

ii) Analysis :- identify inputs, outputs and processing

iii) Algorithm design :- create step by step logic to solve the problem

iv) Flowchart / Pseudocode :- represent the algorithm visually or textually

v) Coding :- convert algorithm into a program using a programming language

vi) Testing & Debugging :- check correctness and remove errors

vii) Documentation & maintenance :- improve and maintain the program

Role of the algorithm :- The algorithm acts as a blueprint of the solution. It ensures correctness, efficiency and makes coding easier and error-free

Q4) Explain linear and non-linear data structures with suitable examples

Ans) → Linear Data Structure :-

Data elements are arranged sequentially

i) Array ii) Stack iii) Queue iv) Linked list

↳ Non-Linear Data Structure :-
Data elements are arranged in a hierarchical or interconnected manner
Example :- Tree, Graph

Q5) Differentiate between algorithm and program. Analyse their roles in SDLC

Ans)

Algorithm	Program
Step by step logic	Written Code
Language independent	Language dependent
Design level	Implementation Level
No syntax rules	Follows syntax rules

- Role in SDLC :-

i) Algorithm :→ used in design phase to plan the solution
ii) Program :→ used in implementation phase to execute the solution on a computer.

Q6) Examine the relationship among data, data structure and algorithm. Justify how the choice of data structure affects algorithm efficiency.

Ans) → Relationship :-
- Data is raw information
- Data structure organizes the data
- Algorithm processes the data using data structures

- Justification :-

The efficiency of an algorithm depends on the data structure used.

↳ Searching in an array → slower [O(n)]
↳ Searching in a binary search tree → faster [O(log n)]

- Conclusion :- choosing the right data structure improves time and space efficiency of algorithms